

Backend Task

We would like to gain a deeper insight into your technical abilities. We have put together a short assignment that we would like you to complete. You are free to spend as much time as you'd like on this, but we did not intend it to take up much of your time; it should be possible to complete it in two hours or so.

Task Description

We would like you to create a Go API. The API should have one route, `/api/v1/zoos/{enclosures}`, where `enclosures` is a number between 1 and 8 representing the number of enclosures that exist at a fictional zoo.

At [this URL](#) you will find a JSON file containing 100 fictional zoo animals. Each animal has a type (e.g. monkey, snake, etc.), popularity (1-10), maximum friends representing how many animals it can share an enclosure with (1-5), and a list of other animal types the animal is incompatible with.¹

When a GET request is made to the above-mentioned API route, it should return a list of animals it would put in each of the enclosures. How you structure that response data is up to you. Each enclosure should have at least one animal, and ideally more than one. Remember, however, to not violate any of the constraints (incompatibility or maximum friends).² And, of course, each animal from the JSON can only be used once.

Please be aware that the main purpose of this task is to see how you structure your code, and how you write APIs. We want you to pretend that this API will have multiple endpoints in the future, as well as multiple developers working on it. The exact mechanism you use for calculating which animals to place in which enclosures is less interesting to us, so please do not get too stuck on that.

Submitting the Task

Once you have completed the task, please make your code available to Kasper Rynning-Tønnesen via whatever means you feel is most convenient and appropriate.³ If you have

¹ Note that the list of incompatible animals is specific to the animal in question, not the animal type; it is possible that one monkey is incompatible with snakes while another monkey is not.

² For bonus points, try to maximize the popularity of the zoo so you find the combination of animals in each enclosure that gives the zoo the highest popularity score (ie sum of the popularity points). Please do not spend forever on this part, though, as we are not sure we even have a good solution here and it is not the main point of this task.

³ Kasper's GitHub account is *kasperrt* if you would like to share the code that way.

any questions, you can also reach out to Kasper Rynning-Tønnesen directly via email at kasper.rynning-tonnesen@pistachioapp.com.